

Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning

July 2022 - Present

Nevada National Security Site

- Developed a YOLO-based object-detection model in PyTorch that achieved a 80% mAP evaluation on a in-house thermal-image vehicle dataset. The model was trained on AWS with four GPUs by leveraging PyTorch's DataParallel.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Took initiative to develop the company's first deep-learning interface. The interface is a wrapper on PyTorch and offers features that include:
 - the introduction of a standardized approach for training, validating and testing models to ensure reproducibility and model validity,
 - the guarantee that code written by all colleagues can seamlessly be integrated,
 - support for multi-GPU training with the use of PyTorch's DataParallel,
 - implementation of dry-run checks to ensure the safe execution of a program.
- Maintainer of my group's AWS EC2 instance. Ensured the health of the instance, creation of user-accounts and GPU functionality.
- Sole developer of a Django web interface, which wrapped Neo4j. The interface gave non-technical staff the ability to query and update the company's Neo4j graph-database without needing to know any query syntax.

Technical Skills

Programming Languages: Python, SQL, Bash, Cypher, Lua, L^AT_EX

Libraries: PyTorch, Scikit-Learn, Pandas, Scipy, Sqlalchemy, Matplotlib, Plotly, Streamlit, Django

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, Neo4j

Education

Auburn University - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Eisenberg, N., Chen, L. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). <https://doi.org/10.1007/s10959-023-01302-4>